

END-OF-CHAPTER EXERCISES

Chapter 8

Predicting What Comes Next: Exploring Sequences and Progressions

RAVI SHANKAR

Mathematics

Key Idea

Chapter 8 – Must Know Formulas

Arithmetic Progression (AP):

$$t_n = a + (n - 1)d$$

Geometric Progression (GP):

$$t_n = ar^{n-1}$$

Sum of first n natural numbers:

$$S_n = \frac{n(n+1)}{2}$$

AP common difference:

$$d = t_2 - t_1$$

GP common ratio:

$$r = \frac{t_2}{t_1}$$

Question 1

An AP has its 11th term equal to 38 and its 16th term equal to 73. Find its 31st term.

Solution

Let the first term be a and common difference be d .

Using

$$t_n = a + (n - 1)d$$

For the 11th term,

$$a + 10d = 38 \quad (1)$$

For the 16th term,

$$a + 15d = 73 \quad (2)$$

Subtracting (1) from (2),

$$5d = 35$$

$$d = 7.$$

From (1),

$$a + 10(7) = 38$$

$$a + 70 = 38$$

$$a = -32.$$

Therefore,

$$\begin{aligned} t_{31} &= a + 30d \\ &= -32 + 30(7) \\ &= -32 + 210 \\ &= \boxed{178}. \end{aligned}$$

Final Answer

$$\boxed{t_{31} = 178}$$

Question 2

Find the AP whose third term is 16 and whose seventh term exceeds the fifth term by 12.

Solution

Let the first term be a and common difference be d .

Third term:

$$a + 2d = 16 \quad (1)$$

Also,

$$\begin{aligned} t_7 - t_5 &= 12 \\ (a + 6d) - (a + 4d) &= 12 \\ 2d &= 12 \\ d &= 6. \end{aligned}$$

Putting $d = 6$ in (1),

$$\begin{aligned} a + 2(6) &= 16 \\ a + 12 &= 16 \\ a &= 4. \end{aligned}$$

Hence the AP is

$$4, 10, 16, 22, 28, \dots$$

Final Answer

$$a = 4, \quad d = 6$$

$$4, 10, 16, 22, 28, \dots$$

Question 3

Find the number of three-digit numbers divisible by 7.

Solution

The smallest three-digit multiple of 7 is

$$105$$

and the largest is

$$994.$$

Thus,

$$105, 112, 119, \dots, 994$$

is an AP.

Here,

$$a = 105, \quad d = 7.$$

Using

$$t_n = a + (n - 1)d$$

we have

$$994 = 105 + (n - 1)7$$

$$889 = 7(n - 1)$$

$$127 = n - 1$$

$$n = 128.$$

Final Answer

128 three-digit numbers

Question 4

Find the number of multiples of 4 lying strictly between 10 and 250.

Solution

The first multiple of 4 greater than 10 is

$$12$$

and the last multiple less than 250 is

$$248.$$

Therefore,

$$12, 16, 20, \dots, 248$$

is an AP.

Here,

$$a = 12, \quad d = 4.$$

Using the last term,

$$248 = 12 + (n - 1)4$$

$$236 = 4(n - 1)$$

$$59 = n - 1$$

$$n = 60.$$

Final Answer

$$\boxed{60}$$

Question 5

Find all possible GPs for which the sum of the first two terms is -4 and the fifth term is four times the third term.

Solution

Let the GP be

$$a, ar, ar^2, ar^3, ar^4, \dots$$

Given,

$$\begin{aligned} a + ar &= -4 \\ a(1 + r) &= -4. \end{aligned} \tag{1}$$

Also,

$$\begin{aligned} ar^4 &= 4ar^2 \\ r^2 &= 4. \end{aligned}$$

Hence,

$$r = 2 \quad \text{or} \quad r = -2.$$

Case 1: $r = 2$

From (1),

$$\begin{aligned} 3a &= -4 \\ a &= -\frac{4}{3}. \end{aligned}$$

Thus,

$$\boxed{-\frac{4}{3}, -\frac{8}{3}, -\frac{16}{3}, -\frac{32}{3}, \dots}$$

Case 2: $r = -2$

From (1),

$$\begin{aligned} a(1 - 2) &= -4 \\ -a &= -4 \\ a &= 4. \end{aligned}$$

Thus,

$$\boxed{4, -8, 16, -32, 64, \dots}$$

Question 6

Express 100 in all possible ways as a sum of consecutive natural numbers.

Solution

One obvious representation is

$$100 = 100.$$

For 5 consecutive numbers, let the middle number be x .

$$(x - 2) + (x - 1) + x + (x + 1) + (x + 2) = 100$$

$$5x = 100$$

$$x = 20.$$

Therefore,

$$100 = 18 + 19 + 20 + 21 + 22.$$

For 8 consecutive numbers, let the first number be x .

$$x + (x + 1) + \cdots + (x + 7) = 100$$

$$8x + 28 = 100$$

$$8x = 72$$

$$x = 9.$$

Therefore,

$$100 = 9 + 10 + 11 + 12 + 13 + 14 + 15 + 16.$$

Final Answer

$$\boxed{100}$$

$$\boxed{18 + 19 + 20 + 21 + 22}$$

$$\boxed{9 + 10 + 11 + 12 + 13 + 14 + 15 + 16}$$

Question 7

A bacterial culture starts with 30 bacteria and doubles every hour. Find the number after the 2nd hour, 4th hour and n th hour.

Solution

The number of bacteria is multiplied by 2 every hour.

Therefore,

$$30, 60, 120, 240, 480, \dots$$

This is a GP with

$$a = 30, \quad r = 2.$$

Using

$$t_n = ar^{n-1},$$

if t_n denotes the number at the n th stage after the initial count, we obtain

$$t_2 = 30(2)^2 = 120,$$

$$t_4 = 30(2)^4 = 480.$$

The number at the end of the n th hour is

$$\boxed{30(2)^n}.$$

Final Answer

$$\boxed{120, \quad 480, \quad 30(2)^n}$$

Question 8

The sum of the 4th and 8th terms of an AP is 24, while the sum of the 6th and 10th terms is 44. Find its first three terms.

Solution

Let the first term be a and common difference be d .

$$\begin{aligned}t_4 + t_8 &= 24 \\(a + 3d) + (a + 7d) &= 24 \\2a + 10d &= 24 \\a + 5d &= 12.\end{aligned}\tag{1}$$

Similarly,

$$\begin{aligned}t_6 + t_{10} &= 44 \\(a + 5d) + (a + 9d) &= 44 \\2a + 14d &= 44 \\a + 7d &= 22.\end{aligned}\tag{2}$$

Subtracting (1) from (2),

$$\begin{aligned}2d &= 10 \\d &= 5.\end{aligned}$$

Using (1),

$$\begin{aligned}a + 5(5) &= 12 \\a &= -13.\end{aligned}$$

Therefore, the first three terms are

$$-13, -8, -3.$$

Final Answer

$$\boxed{-13, -8, -3}$$

Question 9

Find the smallest natural number n for which the sum of the first n natural numbers exceeds 1000.

Solution

We know

$$S_n = \frac{n(n+1)}{2}.$$

We require

$$\begin{aligned}\frac{n(n+1)}{2} &> 1000 \\ n(n+1) &> 2000.\end{aligned}$$

Now,

$$\begin{aligned}S_{44} &= \frac{44(45)}{2} \\ &= 990 < 1000.\end{aligned}$$

But

$$\begin{aligned}S_{45} &= \frac{45(46)}{2} \\ &= 1035 > 1000.\end{aligned}$$

Hence the smallest possible value is

$$\boxed{n = 45}.$$

Question 10

For the GP $2, 8, 32, \dots$, determine which term is 131072. Also write its explicit and recursive rules.

Solution

Here,

$$a = 2, \quad r = \frac{8}{2} = 4.$$

Therefore,

$$t_n = 2(4)^{n-1}$$

We need

$$2(4)^{n-1} = 131072$$

$$4^{n-1} = 65536$$

$$4^{n-1} = 4^8.$$

Hence,

$$n - 1 = 8$$

$$n = 9.$$

The recursive rule is

$$t_1 = 2, \quad t_n = 4t_{n-1}$$

for $n \geq 2$.

Final Answer

131072 is the 9th term.

$$t_n = 2(4)^{n-1}$$

$$t_1 = 2, \quad t_n = 4t_{n-1}$$

Question 11

The first three terms of a GP have sum $\frac{13}{12}$ and product -1 . Find the possible common ratios and the terms.

Solution

Let the three terms be

$$\frac{a}{r}, \quad a, \quad ar.$$

Their product is

$$\frac{a}{r} \cdot a \cdot ar = a^3 = -1.$$

Thus,

$$a = -1.$$

Hence the terms are

$$-\frac{1}{r}, \quad -1, \quad -r.$$

Their sum is

$$-\frac{1}{r} - 1 - r = \frac{13}{12}.$$

Multiplying by $12r$,

$$\begin{aligned} -12 - 12r - 12r^2 &= 13r \\ 12r^2 + 25r + 12 &= 0 \\ (4r + 3)(3r + 4) &= 0. \end{aligned}$$

Therefore,

$$r = -\frac{3}{4} \quad \text{or} \quad r = -\frac{4}{3}.$$

If

$$r = -\frac{3}{4},$$

the terms are

$$\frac{4}{3}, \quad -1, \quad \frac{3}{4}.$$

If

$$r = -\frac{4}{3},$$

the terms are

$$\frac{3}{4}, \quad -1, \quad \frac{4}{3}.$$

Final Answer

$$r = -\frac{3}{4} \text{ or } -\frac{4}{3}$$

$$\frac{4}{3}, -1, \frac{3}{4}$$

or

$$\frac{3}{4}, -1, \frac{4}{3}$$

Question 12

In a GP, the 4th, 10th and 16th terms are respectively x , y and z . Prove that x, y, z are themselves in GP.

Solution

Let the first term be a and common ratio be r .

Then,

$$\begin{aligned}x &= ar^3, \\y &= ar^9, \\z &= ar^{15}.\end{aligned}$$

Now,

$$\begin{aligned}\frac{y}{x} &= \frac{ar^9}{ar^3} \\&= r^6.\end{aligned}$$

Also,

$$\begin{aligned}\frac{z}{y} &= \frac{ar^{15}}{ar^9} \\&= r^6.\end{aligned}$$

Therefore,

$$\frac{y}{x} = \frac{z}{y}.$$

Hence, x, y, z are in GP.

Final Answer x, y, z are in GP.**Question 13**

The sum of the first three terms of a GP is 26, and the sum of their squares is 364. Find the terms.

Solution

Let the three terms be

$$a, ar, ar^2.$$

Given,

$$\begin{aligned} a + ar + ar^2 &= 26 \\ a(1 + r + r^2) &= 26. \end{aligned} \tag{1}$$

Also,

$$\begin{aligned} a^2 + a^2r^2 + a^2r^4 &= 364 \\ a^2(1 + r^2 + r^4) &= 364. \end{aligned} \tag{2}$$

Dividing (2) by the square of (1),

$$\begin{aligned} \frac{1 + r^2 + r^4}{(1 + r + r^2)^2} &= \frac{364}{26^2} \\ &= \frac{7}{13}. \end{aligned}$$

Therefore,

$$13(1 + r^2 + r^4) = 7(1 + r + r^2)^2.$$

Using

$$1 + r^2 + r^4 = (r^2 + r + 1)(r^2 - r + 1),$$

we obtain

$$\begin{aligned} 13(r^2 - r + 1) &= 7(r^2 + r + 1) \\ 6r^2 - 20r + 6 &= 0 \\ 3r^2 - 10r + 3 &= 0 \\ (3r - 1)(r - 3) &= 0. \end{aligned}$$

Hence,

$$r = \frac{1}{3} \quad \text{or} \quad r = 3.$$

For $r = \frac{1}{3}$,

$$\begin{aligned} a \left(1 + \frac{1}{3} + \frac{1}{9} \right) &= 26 \\ a \left(\frac{13}{9} \right) &= 26 \end{aligned}$$

Final Answer

$$18, 6, 2$$

or equivalently,

$$2, 6, 18$$

Question 14

A sequence is defined by

$$P_1 = 1, \quad P_2 = 2$$

and, for $n > 2$,

$$P_n = P_1 + P_2 + \cdots + P_{n-1} + 1.$$

Find P_1 to P_8 , then find simpler recursive and explicit rules.

Solution

We calculate successively:

$$P_1 = 1,$$

$$P_2 = 2,$$

$$P_3 = 1 + 2 + 1 = 4,$$

$$P_4 = 1 + 2 + 4 + 1 = 8,$$

$$P_5 = 1 + 2 + 4 + 8 + 1 = 16,$$

$$P_6 = 32,$$

$$P_7 = 64,$$

$$P_8 = 128.$$

Thus,

$$1, 2, 4, 8, 16, 32, 64, 128.$$

Observe that every term is twice the previous term.

Therefore, a simpler recursive rule is

$$P_1 = 1, \quad P_n = 2P_{n-1}$$

for $n \geq 2$.

Hence the explicit formula is

$$P_n = 2^{n-1}.$$

Final Answer

$$P_1, \dots, P_8 = 1, 2, 4, 8, 16, 32, 64, 128$$

$$P_n = 2P_{n-1}$$

$$P_n = 2^{n-1}$$

Question 15

A sequence is defined by

$$W_1 = 1, \quad W_2 = 2$$

and, for $n > 2$,

$$W_n = W_1 + W_2 + \cdots + W_{n-2} + 2.$$

Find W_1 to W_8 and identify the sequence.

Solution

Calculate successively:

$$\begin{aligned}
 W_1 &= 1, \\
 W_2 &= 2, \\
 W_3 &= W_1 + 2 = 1 + 2 = 3, \\
 W_4 &= W_1 + W_2 + 2 \\
 &= 1 + 2 + 2 = 5, \\
 W_5 &= 1 + 2 + 3 + 2 = 8, \\
 W_6 &= 1 + 2 + 3 + 5 + 2 = 13, \\
 W_7 &= 1 + 2 + 3 + 5 + 8 + 2 = 21, \\
 W_8 &= 1 + 2 + 3 + 5 + 8 + 13 + 2 = 34.
 \end{aligned}$$

Therefore,

$$\boxed{1, 2, 3, 5, 8, 13, 21, 34}.$$

Notice that after the first two terms, every term is the sum of the two preceding terms:

$$\begin{aligned}
 1 + 2 &= 3, \\
 2 + 3 &= 5, \\
 3 + 5 &= 8, \\
 5 + 8 &= 13.
 \end{aligned}$$

This is the famous

Virahanka–Fibonacci sequence.

Its simpler recursive rule is

$$\boxed{W_1 = 1, \quad W_2 = 2, \quad W_n = W_{n-1} + W_{n-2}}$$

for $n \geq 3$.

Final Answer

$$\boxed{1, 2, 3, 5, 8, 13, 21, 34}$$

Virahanka–Fibonacci sequence

Important**CHAPTER 8 – FINAL REVISION**
$$\text{AP} \longrightarrow \text{Same Difference}$$

$$t_n = a + (n - 1)d$$

$$\text{GP} \longrightarrow \text{Same Ratio}$$

$$t_n = ar^{n-1}$$

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

Remember:

- AP **difference** check
- GP **ratio** check
- Explicit rule n value put
- Recursive rule previous terms
- Sequence pattern step

Think $\rightarrow$ Find the Pattern $\rightarrow$ Choose the Rule**RAVI SHANKAR – Mathematics***“Mathematics becomes easy when the pattern becomes clear.”*